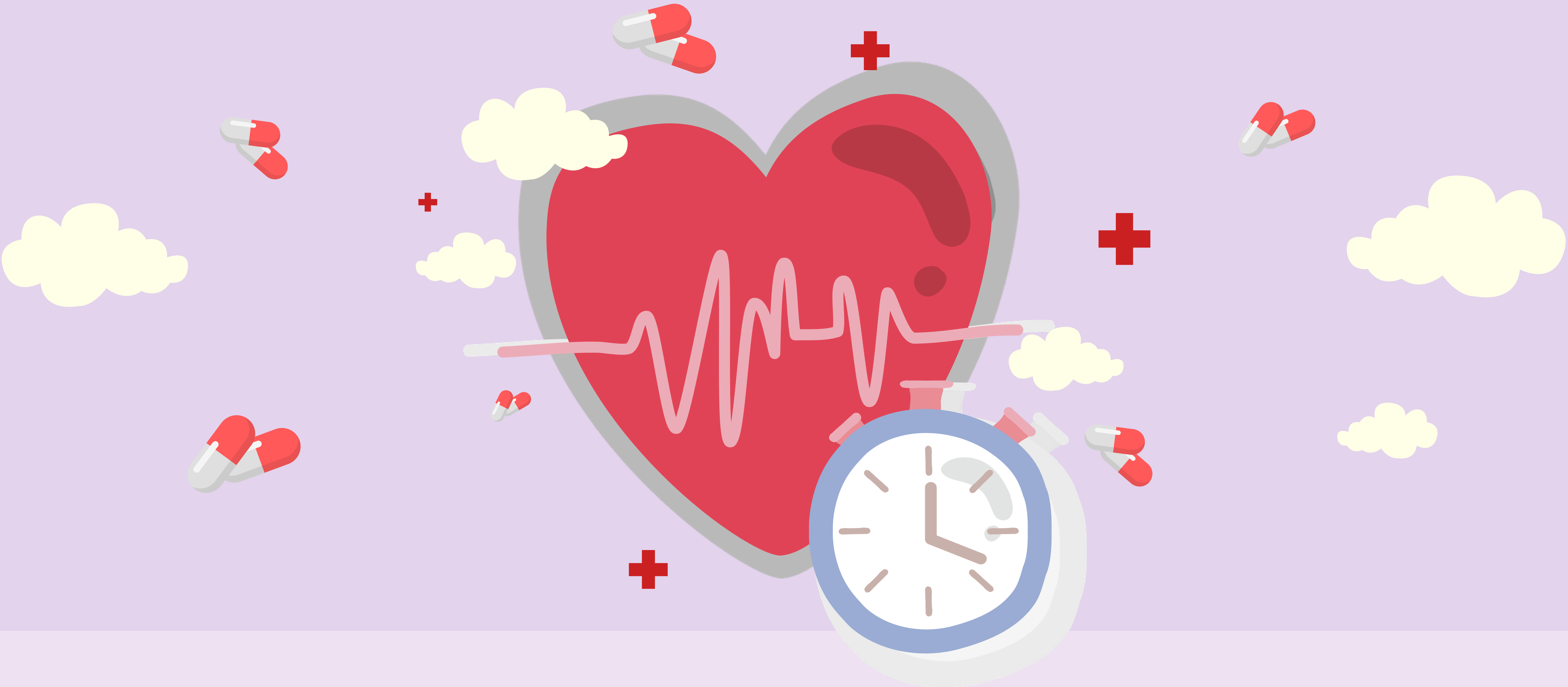


CARDIAC ARREST PREDICTION & DETECTION

COMPREHENSIVE LITERATURE ANALYSIS OF EXISTING RESEARCH MODELS, ALGORITHMS, RISK FACTORS AND HEALTHCARE MONITORING SYSTEMS

"WHY CARDIAC ARREST PREDICTION MATTERS ?"

- Cardiovascular diseases are leading cause of global mortality
- Sudden cardiac arrest occurs without early visible symptoms
- Survival depends heavily on early detection
- Traditional diagnosis is:
  - Reactive (after symptoms)
  - Not continuous
- Need for predictive + real-time AI systems

Key Points:

- High-risk groups: athletes, elderly, diabetic patients
- Lifestyle factors increasing risk:
  - Stress, obesity, inactivity

BLOOD PRESSURE CATEGORIES

According to standard guidelines, blood pressure levels are categorized as:

Normal

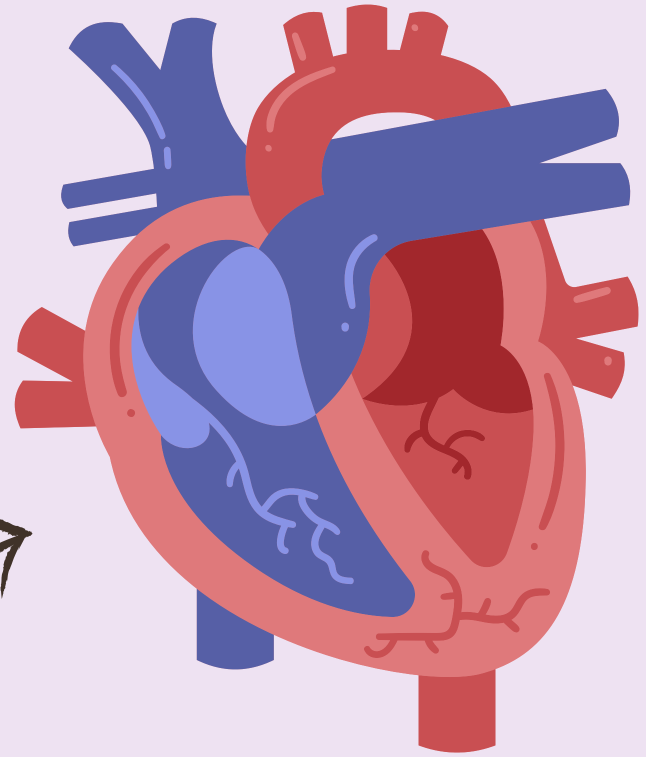
Less than 120/80 mmHg

Elevated

Systolic 120–129 and diastolic less than 80 mmHg

Hypertension Stage 1

Systolic 130–139 or diastolic 80–89 mmHg



Hypertension Stage 2

Systolic 140 or higher or diastolic 90 or higher

Hypertensive Crisis

Systolic over 180 and/or diastolic over 120 mmHg

ANALYSIS ON EXISTING RESEARCH PAPERS

Multiple studies focus on:

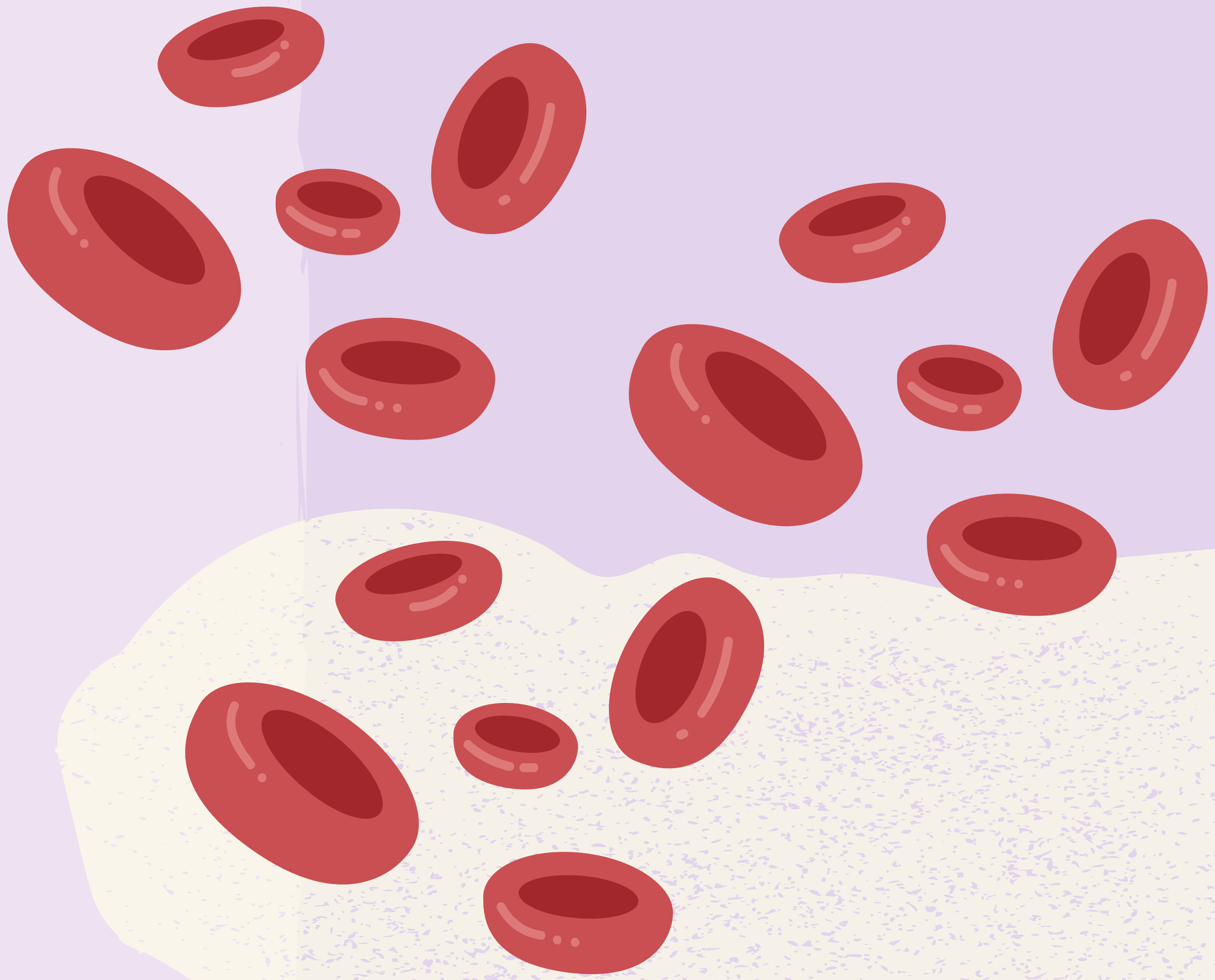
- Heart disease prediction
- Arrhythmia detection
- ECG-based anomaly detection

Common Approaches Identified:

- Machine Learning models for risk prediction
- Deep Learning for ECG signal analysis
- IoT-based monitoring systems

Key Observation:

- Accuracy is high in controlled datasets
- Real-time implementation is still limited



ALGORITHMS USED

Machine Learning & Deep Learning Models

Traditional ML:

- Logistic Regression
- Decision Tree
- Random Forest
- Support Vector Machine (SVM)
- K-Nearest Neighbors (KNN)

Deep Learning:

- Artificial Neural Networks (ANN)
- Convolutional Neural Networks (CNN) → ECG
- LSTM → Time-series prediction

Performance Insight

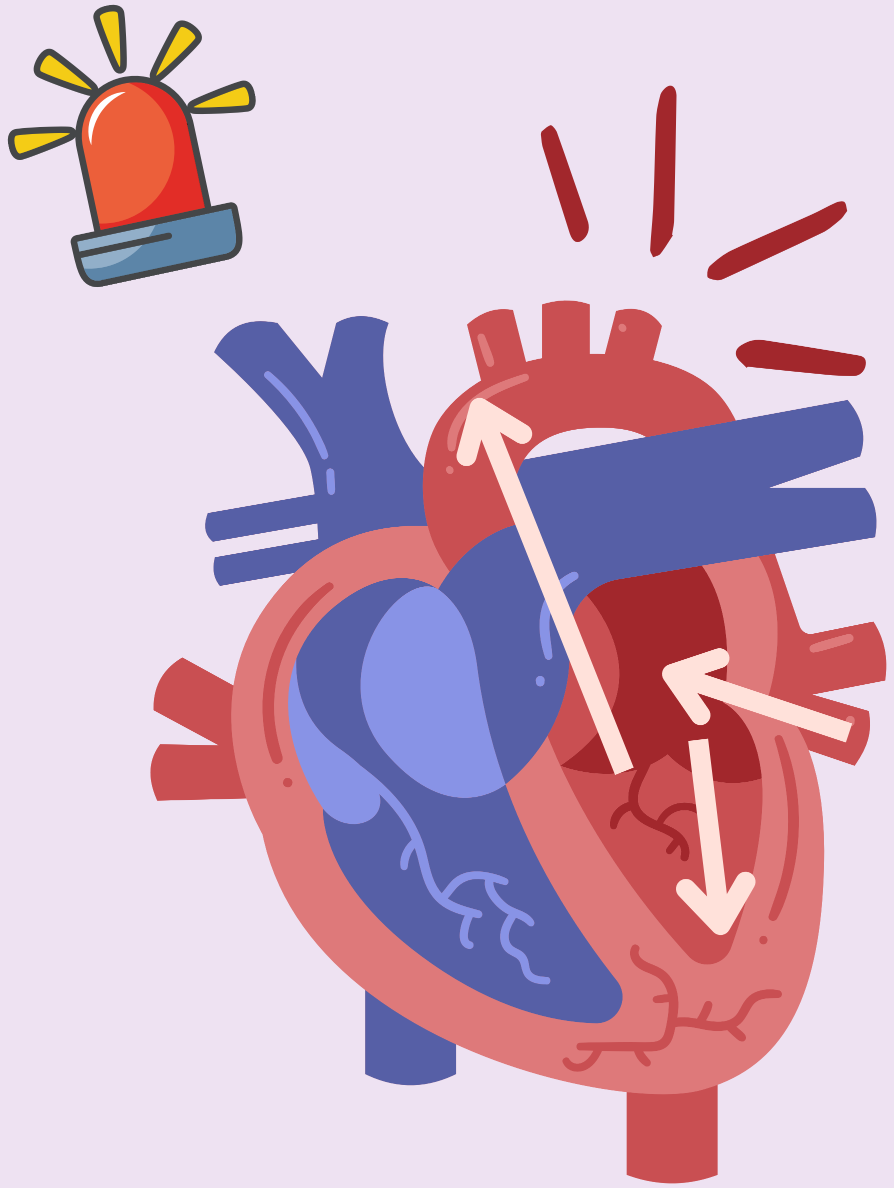
- Decision Tree → ~98% accuracy
- CNN → ~97% accuracy
- DNN → ~94–95% accuracy

Conclusion:

- Deep learning performs better for ECG
- ML models good for structured data

EXISTING SYSTEM LIMITATIONS

Challenges Identified from Research



1

Technical Limitations:

Use of static datasets

Lack of real-time monitoring

Poor generalization

2

Practical Issues:

High implementation cost

Not wearable-friendly

Limited accessibility

3

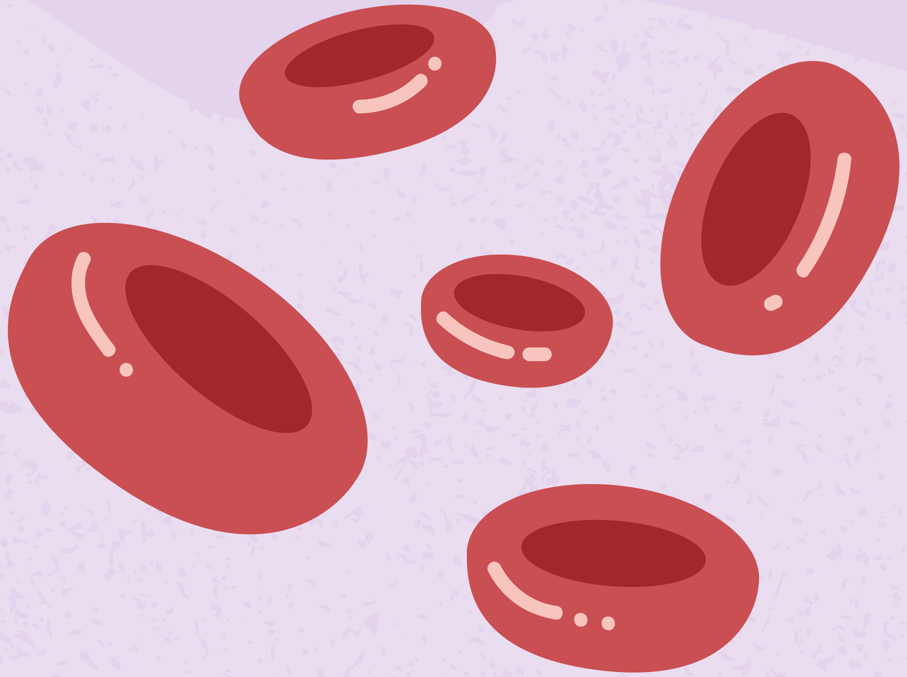
Model Issues:

Black-box models (low explainability)

False positives

PROPOSED DIRECTION

Future System Vision



Content:

Real-time cardiac monitoring system
AI-based risk prediction
Continuous ECG signal analysis
IoT + wearable integration

Features:

Live health tracking
Early warning alerts
Cloud-based monitoring
Doctor dashboard

Innovation:

Hybrid ML + IoT system
Cost-effective solution
Personalized prediction

COST & CONCLUSION

The future of cardiac healthcare lies in predictive, preventive, and AI-driven intelligent monitoring systems.

Hardware Components:
ECG Sensor
Pulse Sensor
Wearable Device Integration

Software Components:
Machine Learning Models
Cloud Storage (Firebase / MongoDB)
Web Dashboard (React)

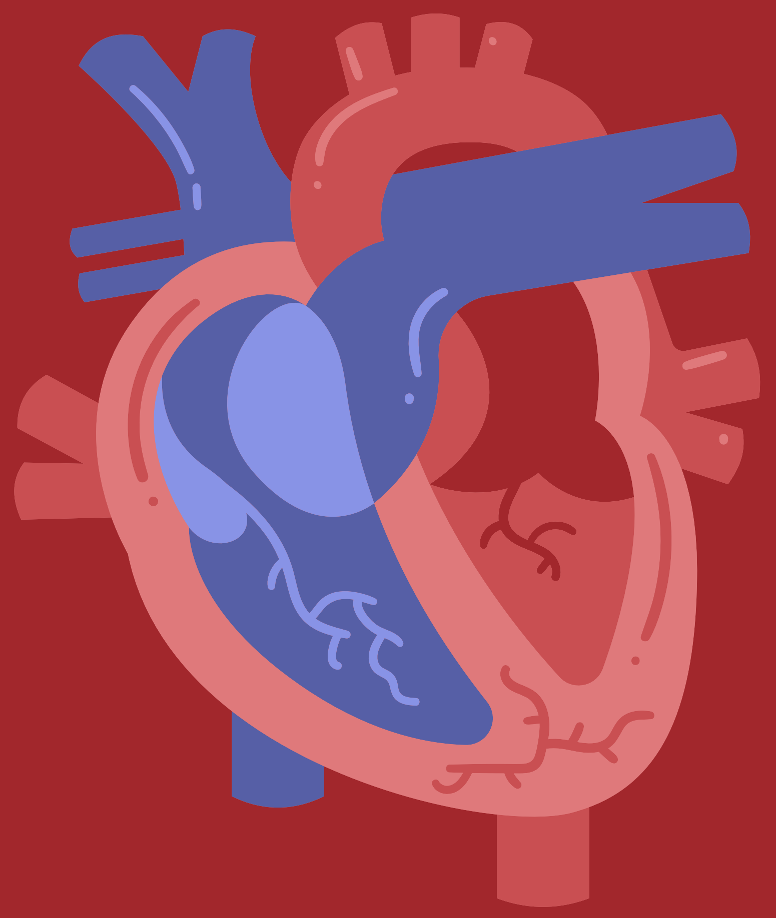
Deployment Cost:
Server hosting
Data processing
Maintenance

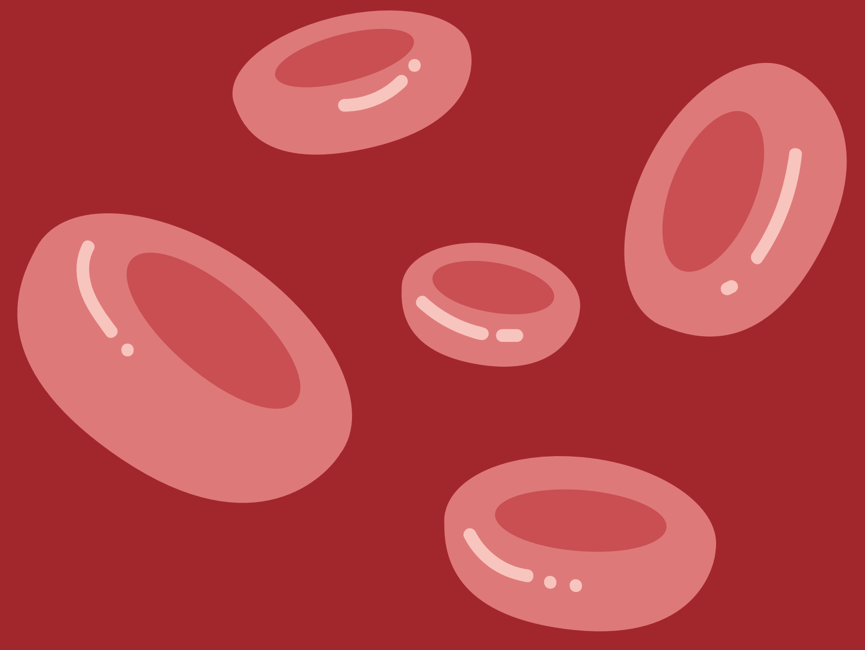
Conclusion

Existing research proves ML is effective for cardiac prediction
Deep learning improves ECG-based detection accuracy
Most current systems lack real-time monitoring & affordability

TIP Take medication consistently and check blood pressure regularly to ensure effectiveness.

THE CIRCULATORY SYSTEM





The role of the heart, blood and blood vessels

THANK YOU

Do you have any questions?